Java Zero to Hero

Practice Questions, Algorithms & Comprehensive Project Workbook

COMPLETE STUDY & PRACTICE GUIDE

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1. Conditional Statements – Practice Questions

IF Statement

`Question 1` **Check Whether a Number is Positive**

Problem Statement: Write a Java program to read an integer and determine whether it is a positive number. If the number is greater than zero, print "Positive Number". Otherwise, print nothing.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Positive Number" if N > 0.

SAMPLE INPUT

```
15
```

SAMPLE OUTPUT

```
Positive Number
```

Question 2 Check Whether a String is Empty

Problem Statement: Write a Java program to read a string and determine whether it is empty. Print an appropriate message based on the result.

INPUT FORMAT

A string.

OUTPUT FORMAT

Print "String is Empty" or "String is Not Empty".

SAMPLE INPUT

```
""
```

SAMPLE OUTPUT

```
String is Empty
```

Question 3 Determine Whether a Number is Positive, Negative, or Zero (Using Only if Statements)

Problem Statement: Write a Java program to determine whether a given integer is positive, negative, or zero using only "if" statements (without using "else if" or "else").

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Positive", "Negative", or "Zero".

SAMPLE INPUT

```
-10
```

SAMPLE OUTPUT

```
Negative
```

Question 4 — Check Whether a Number is Divisible by Both 2 and 3

Problem Statement: Write a Java program to determine whether the given number is divisible by both 2 and 3.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Divisible" or "Not Divisible".

SAMPLE INPUT

```
24
```

SAMPLE OUTPUT

```
Divisible
```

Question 5 — Check Whether a Number is Divisible by Both 3 and 5

Problem Statement: Write a Java program to determine whether the given number is divisible by both 3 and 5.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Divisible" or "Not Divisible".

SAMPLE INPUT

```
45
```

SAMPLE OUTPUT

```
Divisible
```

Question 6 Determine Whether a Number is a Perfect Square

Problem Statement: Write a Java program to determine whether the given number is a perfect square.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Perfect Square" or "Not a Perfect Square".

SAMPLE INPUT

```
64
```

SAMPLE OUTPUT

```
Perfect Square
```

Question 7 Determine Whether a Number is a Perfect Cube

Problem Statement: Write a Java program to determine whether the given number is a perfect cube.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Perfect Cube" or "Not a Perfect Cube".

SAMPLE INPUT

```
125
```

SAMPLE OUTPUT

```
Perfect Cube
```

`Question 8`

Check Whether a Number is a Multiple of 4

Problem Statement: Write a Java program to determine whether the given number is a multiple of 4.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Multiple of 4" or "Not a Multiple of 4".

SAMPLE INPUT

```
20
```

SAMPLE OUTPUT

```
Multiple of 4
```

IF-ELSE Statement

`Question 9`

Check Whether a Year is a Leap Year

Problem Statement: Write a Java program to determine whether a given year is a leap year according to the Gregorian calendar.

INPUT FORMAT

An integer representing the year.

OUTPUT FORMAT

Print "Leap Year" or "Not a Leap Year".

SAMPLE INPUT

```
2024
```

SAMPLE OUTPUT

```
Leap Year
```

Question 10 Check Whether a Year is a Century Year

Problem Statement: Write a Java program to determine whether the given year is a century year.

INPUT FORMAT

An integer representing the year.

OUTPUT FORMAT

Print "Century Year" or "Not a Century Year".

SAMPLE INPUT

```
1900
```

SAMPLE OUTPUT

```
Century Year
```

Question 11 Check Whether a Number is Prime

Problem Statement: Write a Java program to determine whether a given integer is a prime number.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print "Prime Number" or "Not a Prime Number".

SAMPLE INPUT

```
17
```

SAMPLE OUTPUT

```
Prime Number
```

Question 12 Check Voting Eligibility

Problem Statement: Write a Java program to determine whether a person is eligible to vote based on age. A person is eligible if the age is 18 years or above.

INPUT FORMAT

An integer representing age.

OUTPUT FORMAT

Print "Eligible to Vote" or "Not Eligible to Vote".

SAMPLE INPUT

```
20
```

SAMPLE OUTPUT

```
Eligible to Vote
```

Question 13 Check Whether a Number is Positive or Non-Positive

Problem Statement: Write a Java program to determine whether a given integer is positive or non-positive.

INPUT FORMAT

A single integer.

OUTPUT FORMAT

Print "Positive" or "Non-Positive".

SAMPLE INPUT

```
0
```

SAMPLE OUTPUT

```
Non-Positive
```

Question 14 Find the Larger of Two Numbers

Problem Statement: Write a Java program to read two integers and print the larger number.

INPUT FORMAT

Two integers.

OUTPUT FORMAT

Print the largest number.

SAMPLE INPUT

```
15 28
```

SAMPLE OUTPUT

```
28
```

Question 15 Find the Smallest of Three Numbers

Problem Statement: Write a Java program to determine the smallest among three integers without using built-in methods.

INPUT FORMAT

Three integers.

OUTPUT FORMAT

Print the smallest number.

SAMPLE INPUT

```
9 2 6
```

SAMPLE OUTPUT

```
2
```

Question 16 — Find the Largest of Four Numbers

Problem Statement: Write a Java program to determine the largest among four integers.

INPUT FORMAT

Four integers.

OUTPUT FORMAT

Print the largest number.

SAMPLE INPUT

```
12 45 8 30
```

SAMPLE OUTPUT

```
45
```

Question 17 — Check Whether a Character is an Alphabet

Problem Statement: Write a Java program to determine whether a given character is an alphabet.

INPUT FORMAT

A single character.

OUTPUT FORMAT

Print "Alphabet" or "Not an Alphabet".

SAMPLE INPUT

```
A
```

SAMPLE OUTPUT

```
Alphabet
```

Question 18 Check Whether a Character is a Vowel or Consonant

Problem Statement: Write a Java program to determine whether an alphabet is a vowel or a consonant.

INPUT FORMAT

A single alphabet character.

OUTPUT FORMAT

Print "Vowel" or "Consonant".

SAMPLE INPUT

```
e
```

SAMPLE OUTPUT

```
Vowel
```

Question 19 Check Whether a String is a Palindrome

Problem Statement: Write a Java program to determine whether a given string reads the same forwards and backwards.

INPUT FORMAT

A string.

OUTPUT FORMAT

Print "Palindrome" or "Not a Palindrome".

SAMPLE INPUT

```
madam
```

SAMPLE OUTPUT

```
Palindrome
```

Question 20 Find the Absolute Value of a Number

Problem Statement: Write a Java program to print the absolute value of a given integer.

INPUT FORMAT

A single integer.

OUTPUT FORMAT

Print the absolute value.

SAMPLE INPUT

```
-35
```

SAMPLE OUTPUT

```
35
```

IF-ELSE-IF Ladder

Question 21 Temperature Converter

Problem Statement: Convert temperature from Celsius to Fahrenheit or Fahrenheit to Celsius based on the user's choice.

SAMPLE INPUT

```
1
25
```

SAMPLE OUTPUT

```
77.0
```

`Question 22` **Simple Calculator**

Problem Statement: Perform addition, subtraction, multiplication, or division based on the user's selected operator.

SAMPLE INPUT

```
10
5
*
```

SAMPLE OUTPUT

```
50
```

`Question 23` **Grade Calculator**

Problem Statement: Read marks (0–100), validate the input, and print the corresponding grade.

SAMPLE INPUT

```
88
```

SAMPLE OUTPUT

```
Grade A
```

`Question 24` **Days in a Month**

Problem Statement: Read a month number (and year if required) and display the number of days.

SAMPLE INPUT

```
2
2024
```

SAMPLE OUTPUT

```
29
```

Question 25 BMI Calculator

Problem Statement: Calculate Body Mass Index and classify it as Underweight, Normal, Overweight, or Obese.

SAMPLE INPUT

```
70
1.75
```

SAMPLE OUTPUT

```
Normal Weight
```

Question 26 Letter Grade to GPA Converter

Problem Statement: Convert grades A, B, C, D, and F into GPA values.

SAMPLE INPUT

```
A
```

SAMPLE OUTPUT

```
4.0
```

Question 27 Password Strength Checker

Problem Statement: Determine whether a password is Weak, Medium, or Strong based on predefined rules.

SAMPLE INPUT

```
Hello@123
```

SAMPLE OUTPUT

```
Strong Password
```

Question 28 **Purchase Discount Calculator**

Problem Statement: Calculate the final bill after applying discounts based on the purchase amount.

SAMPLE INPUT

```
3500
```

SAMPLE OUTPUT

```
Final Amount = 3150
```

NESTED IF Statement

Question 29 **Even/Odd and Greater Than 10**

Problem Statement: Determine whether a number is even or odd. If it is even, check whether it is greater than 10.

SAMPLE INPUT

```
18
```

SAMPLE OUTPUT

```
Even
Greater than 10
```

Question 30 Number Analysis

Problem Statement: Determine whether a number is positive, negative, or zero. Then determine whether it is even or odd.

SAMPLE INPUT

```
-8
```

SAMPLE OUTPUT

```
Negative
Even
```

Question 31 Voting Eligibility Based on Age and Citizenship

Problem Statement: Determine whether a person is eligible to vote based on age and citizenship.

SAMPLE INPUT

```
20
Yes
```

SAMPLE OUTPUT

```
Eligible to Vote
```

Question 32 Driving Eligibility

Problem Statement: Write a Java program to determine whether a person is eligible to drive a motor vehicle based on their age and possession of a valid driving license. A person is eligible to drive only if they are 18 years of age or older and possess a valid driving license.

INPUT FORMAT

Line 1: Integer age.
Line 2: String ("Yes" or "No").

OUTPUT FORMAT

"Eligible to Drive" or "Not Eligible to Drive".

SAMPLE INPUT 1

```
20
Yes
```

SAMPLE OUTPUT 1

```
Eligible to Drive
```

SAMPLE INPUT 2

```
17
Yes
```

SAMPLE OUTPUT 2

```
Not Eligible to Drive
```

Question 33

Credit Card Eligibility

Problem Statement: Write a Java program to determine whether a person is eligible to apply for a credit card. A person is considered eligible if they are 21 years of age or older and have an annual income of ₹3,00,000 or more.

INPUT FORMAT

Line 1: Integer age.
Line 2: Double annual income.

OUTPUT FORMAT

"Eligible for Credit Card" or "Not Eligible for Credit Card".

SAMPLE INPUT 1

```
25
450000
```

SAMPLE OUTPUT 1

```
Eligible for Credit Card
```

SAMPLE INPUT 2

```
19
500000
```

SAMPLE OUTPUT 2

```
Not Eligible for Credit Card
```

Question 34 Arrange Three Numbers in Ascending Order

Problem Statement: Write a Java program to read three integers and arrange them in ascending order using nested "if" statements. Do not use arrays, loops, or built-in sorting methods.

INPUT FORMAT

A single line containing three integers separated by spaces.

OUTPUT FORMAT

Print the three numbers in ascending order.

SAMPLE INPUT

```
15 7 10
```

SAMPLE OUTPUT

```
7 10 15
```

Question 35 Character Classification

Problem Statement: Write a Java program to classify a given character. If the character is an alphabet, determine whether it is a vowel or a consonant. If it is not an alphabet, print Neither Vowel nor Consonant.

INPUT FORMAT

A single character.

OUTPUT FORMAT

"Vowel", "Consonant", or "Neither Vowel nor Consonant".

SAMPLE INPUT 1

```
A
```

SAMPLE OUTPUT 1

```
Vowel
```

SAMPLE INPUT 2

```
9
```

SAMPLE OUTPUT 2

```
Neither Vowel nor Consonant
```

Question 36

Theme Park Ticket Price

Problem Statement: Write a Java program to calculate the ticket price for a theme park based on the visitor's age and height.
Rules:
- Children below 5 years: Free Entry.
- Visitors aged 5–12 years: Height < 120 cm → ₹150; Height ≥ 120 cm → ₹200
- Visitors aged 13 years and above: Height < 120 cm → ₹250; Height ≥ 120 cm → ₹350

INPUT FORMAT

Line 1: Integer age.
Line 2: Double height (cm).

OUTPUT FORMAT

Print the ticket price.

SAMPLE INPUT

```
15
140
```

SAMPLE OUTPUT

```
Ticket Price = ₹350
```

Question 37 Membership Discount Calculator

Problem Statement: Write a Java program to calculate the final purchase amount based on purchase value and membership status.

Rules:

- If member: Amount ≥ ₹5000 → 20% discount; Amount ≥ ₹2000 < ₹5000 → 10% discount; Otherwise → 5% discount.
- If not member: Amount ≥ ₹5000 → 10% discount; Otherwise → No discount.

INPUT FORMAT

Line 1: Purchase amount.
Line 2: "Yes" or "No" membership.

OUTPUT FORMAT

Print final payable amount.

SAMPLE INPUT

```
6000
Yes
```

SAMPLE OUTPUT

```
Final Amount = ₹4800.00
```

`Question 38` **Scholarship Eligibility**

Problem Statement: Write a Java program to determine whether a student is eligible for a scholarship based on GPA (≥ 8.5) and extracurricular participation ("Yes").

INPUT FORMAT

Line 1: Floating-point GPA.
Line 2: "Yes" or "No".

OUTPUT FORMAT

"Scholarship Approved" or "Scholarship Not Approved".

SAMPLE INPUT 1

```
9.1
Yes
```

SAMPLE OUTPUT 1

```
Scholarship Approved
```

SAMPLE INPUT 2

```
8.9
No
```

SAMPLE OUTPUT 2

```
Scholarship Not Approved
```

2. WHILE Loop – Practice Questions

Question 1 Print Numbers from 1 to 10

Problem Statement: Write a Java program to print the numbers from 1 to 10 using a "while" loop.

INPUT FORMAT

No input.

OUTPUT FORMAT

Print numbers 1 to 10, each on a new line.

SAMPLE OUTPUT

```
1
2
3
4
5
6
7
8
9
10
```

Question 2 Print Numbers in Reverse Order

Problem Statement: Write a Java program to read an integer N and print all numbers from N down to 1 using a "while" loop.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print numbers N to 1, space-separated.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
5 4 3 2 1
```

`Question 3` **Print Even Numbers from 1 to 50**

Problem Statement: Write a Java program to print all even numbers between 1 and 50 using a "while" loop.

INPUT FORMAT

No input.

OUTPUT FORMAT

Print all even numbers from 2 to 50 separated by spaces.

SAMPLE OUTPUT

```
2 4 6 8 10 12 14 16 18 20 22 24 26
28 30 32 34 36 38 40 42 44 46 48 50
```

`Question 4` Multiplication Table

Problem Statement: Write a Java program to read an integer and generate its multiplication table up to 10 using a "while" loop.

INPUT FORMAT

A single integer "N".

OUTPUT FORMAT

Print multiplication table from "N × 1" to "N × 10".

SAMPLE INPUT

```
7
```

SAMPLE OUTPUT

```
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

`Question 5` Factorial of a Number

Problem Statement: Write a Java program to read a positive integer and calculate its factorial using a "while" loop.

INPUT FORMAT

A positive integer "N".

OUTPUT FORMAT

Print the factorial of the given number.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
120
```

Question 6 Sum of Even Numbers

Problem Statement: Write a Java program to calculate the sum of all even numbers between 1 and 100 using a "while" loop.

INPUT FORMAT

No input.

OUTPUT FORMAT

Print the sum of all even numbers from 1 to 100.

SAMPLE OUTPUT

```
2550
```

Question 7 Find All Factors of a Number

Problem Statement: Write a Java program to read an integer and print all of its factors using a "while" loop.

INPUT FORMAT

A positive integer "N".

OUTPUT FORMAT

Print all factors in ascending order.

SAMPLE INPUT

```
12
```

SAMPLE OUTPUT

```
1 2 3 4 6 12
```

`Question 8`

Reverse a Number

Problem Statement: Write a Java program to read an integer and print its reverse using a "while" loop.

INPUT FORMAT	OUTPUT FORMAT
A positive integer "N".	Print the reversed number.

SAMPLE INPUT	SAMPLE OUTPUT
`12345`	`54321`

`Question 9`

Sum of Digits

Problem Statement: Write a Java program to read an integer and calculate the sum of its digits using a "while" loop.

INPUT FORMAT	OUTPUT FORMAT
A positive integer "N".	Print the sum of all digits.

SAMPLE INPUT	SAMPLE OUTPUT
`456`	`15`

Question 10 Greatest Common Divisor (GCD)

Problem Statement: Write a Java program to read two positive integers and find their Greatest Common Divisor (GCD) using the Euclidean algorithm and a "while" loop.

INPUT FORMAT

Two integers separated by a space.

OUTPUT FORMAT

Print the GCD of the two numbers.

SAMPLE INPUT

```
48 18
```

SAMPLE OUTPUT

```
6
```

Question 11 Calculate the Power of a Number

Problem Statement: Write a Java program to read a base and an exponent and calculate $base^{exponent}$ using a "while" loop. Do not use Math.pow().

INPUT FORMAT

Two integers: "base" and "exponent".

OUTPUT FORMAT

Print the calculated power.

SAMPLE INPUT

```
2 5
```

SAMPLE OUTPUT

```
32
```

Question 12 Prime Factorization

Problem Statement: Write a Java program to read a positive integer and print its prime factors using a "while" loop.

INPUT FORMAT

A positive integer "N".

OUTPUT FORMAT

Print all prime factors in ascending order.

SAMPLE INPUT

```
84
```

SAMPLE OUTPUT

```
2 2 3 7
```

Question 13 Fibonacci Series

Problem Statement: Write a Java program to read an integer N and print the first N terms of the Fibonacci sequence using a "while" loop.

INPUT FORMAT

A positive integer "N".

OUTPUT FORMAT

Print the first "N" Fibonacci numbers separated by spaces.

SAMPLE INPUT

```
8
```

SAMPLE OUTPUT

```
0 1 1 2 3 5 8 13
```

Question 14 Prime Numbers from 1 to 100

Problem Statement: Write a Java program to print all prime numbers between 1 and 100 using a "while" loop.

INPUT FORMAT

No input.

OUTPUT FORMAT

Print all prime numbers from 1 to 100 separated by spaces.

SAMPLE OUTPUT

```
2 3 5 7 11 13 17 19 23 29 31 37 41
43 47 53 59 61 67 71 73 79 83 89 97
```

Question 15 Check Whether a String is a Palindrome

Problem Statement: Write a Java program to read a string and determine whether it is a palindrome using a "while" loop.

INPUT FORMAT

A string.

OUTPUT FORMAT

"Palindrome" or "Not a Palindrome".

SAMPLE INPUT

```
madam
```

SAMPLE OUTPUT

```
Palindrome
```

`Question 16` **Reverse a String**

Problem Statement: Write a Java program to read a string and print its reverse using a "while" loop and the StringBuilder class.

INPUT FORMAT

A string.

OUTPUT FORMAT

Print the reversed string.

SAMPLE INPUT

```
Java
```

SAMPLE OUTPUT

```
avaJ
```

`Question 17` **Search for a Word in a String**

Problem Statement: Write a Java program to read a sentence and a word, then use a "while" loop to find the index of the first occurrence of the given word. If not found, print -1.

INPUT FORMAT

Line 1: Sentence.
Line 2: Word to search.

OUTPUT FORMAT

Index of occurrence or "-1".

SAMPLE INPUT

```
Java is a programming language
programming
```

SAMPLE OUTPUT

```
10
```

`Question 18` **Binary to Decimal Conversion**

Problem Statement: Write a Java program to read a binary number and convert it into its decimal equivalent using a "while" loop.

INPUT FORMAT

A binary number.

OUTPUT FORMAT

Print the decimal equivalent.

SAMPLE INPUT

```
101101
```

SAMPLE OUTPUT

```
45
```

3. FOR Loop – Practice Questions

`Question 1` **Print Numbers from 1 to 5**

Problem Statement: Write a Java program to print numbers from 1 to 5 using a "for" loop.

SAMPLE OUTPUT

```
1
2
3
4
5
```

`Question 2` Print Even Numbers from 1 to 10

Problem Statement: Write a Java program to print all even numbers between 1 and 10 using a "for" loop.

SAMPLE OUTPUT

```
2 4 6 8 10
```

`Question 3` Print a Countdown from a Given Number

Problem Statement: Write a Java program to read an integer N and print the numbers from N down to 1 using a "for" loop.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
5 4 3 2 1
```

`Question 4`

Generate the Multiplication Table

Problem Statement: Write a Java program to read an integer and generate its multiplication table up to 10 using a "for" loop.

SAMPLE INPUT

```
8
```

SAMPLE OUTPUT

```
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
8 x 9 = 72
8 x 10 = 80
```

`Question 5`

Generate Multiples of a Number

Problem Statement: Write a Java program to read two integers, N and R, and print all multiples of N from 1 to R using a "for" loop.

SAMPLE INPUT

```
4
25
```

SAMPLE OUTPUT

```
4 8 12 16 20 24
```

`Question 6` **Calculate the Sum of Numbers from 1 to 10**

Problem Statement: Write a Java program to calculate the sum of all integers from 1 to 10 using a "for" loop.

SAMPLE OUTPUT

```
55
```

`Question 7` **Calculate the Sum of the First N Natural Numbers**

Problem Statement: Write a Java program to read an integer N and calculate the sum of the first N natural numbers using a "for" loop.

SAMPLE INPUT

```
10
```

SAMPLE OUTPUT

```
55
```

`Question 8` **Calculate the Sum of Odd Numbers from 1 to 100**

Problem Statement: Write a Java program to calculate the sum of all odd numbers between 1 and 100 using a "for" loop.

SAMPLE OUTPUT

```
2500
```

Question 9 Calculate the Sum of Even Numbers from 1 to 100

Problem Statement: Write a Java program to calculate the sum of all even numbers between 1 and 100 using a "for" loop.

SAMPLE OUTPUT

```
2550
```

Question 10 Calculate the Factorial of a Number

Problem Statement: Write a Java program to read a positive integer and calculate its factorial using a "for" loop.

SAMPLE INPUT

```
6
```

SAMPLE OUTPUT

```
720
```

Question 11 Calculate the Power of a Number

Problem Statement: Write a Java program to read base and exponent and calculate $base^{exponent}$ using a "for" loop. Do not use Math.pow().

SAMPLE INPUT

```
3 4
```

SAMPLE OUTPUT

```
81
```

Question 12 Generate Prime Numbers Within a Given Range

Problem Statement: Write a Java program to read starting and ending range values and print all prime numbers within that range using a "for" loop.

SAMPLE INPUT

```
10 30
```

SAMPLE OUTPUT

```
11 13 17 19 23 29
```

Question 13 Count Prime Numbers in a Given Range

Problem Statement: Write a Java program to read a range and count how many prime numbers are present within that range using a "for" loop.

SAMPLE INPUT

```
1 20
```

SAMPLE OUTPUT

```
8
```

Question 14 Calculate the Sum of Digits of a Number

Problem Statement: Write a Java program to read an integer and calculate the sum of its digits using a "for" loop.

SAMPLE INPUT

```
5678
```

SAMPLE OUTPUT

```
26
```

`Question 15`

Calculate the Least Common Multiple (LCM) of Two Numbers

Problem Statement: Write a Java program to read two positive integers and determine their Least Common Multiple (LCM) using a "for" loop.

SAMPLE INPUT

```
12 18
```

SAMPLE OUTPUT

```
36
```

`Question 16`

Print Elements of an Array

Problem Statement: Write a Java program to read the elements of an integer array and print each element using a "for" loop.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
10 20 30 40 50
```

`Question 17`

Find the Largest Element in an Array

Problem Statement: Write a Java program to read an array of integers and find the largest element using a "for" loop.

SAMPLE INPUT

```
6
12 45 8 67 23 19
```

SAMPLE OUTPUT

```
67
```

Question 18 Calculate the Average of Array Elements

Problem Statement: Write a Java program to read an array of integers and calculate the average of its elements using a "for" loop.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
30.0
```

Question 19 Calculate the Product of Array Elements

Problem Statement: Write a Java program to read an array of integers and calculate the product of all its elements using a "for" loop.

SAMPLE INPUT

```
4
2 3 4 5
```

SAMPLE OUTPUT

```
120
```

Question 20 Reverse an Array

Problem Statement: Write a Java program to read an array of integers and print its elements in reverse order using a "for" loop.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
50 40 30 20 10
```

Question 21 Find the Index of an Element

Problem Statement: Write a Java program to read an array of integers and a target element. Search for the target element using a "for" loop and print its index. If not found, print -1.

SAMPLE INPUT

```
5
15 25 35 45 55
35
```

SAMPLE OUTPUT

```
2
```

Question 22 Find Common Elements Between Two Arrays

Problem Statement: Write a Java program to read two integer arrays and print the common elements using nested "for" loops.

SAMPLE INPUT

```
5
1 2 3 4 5
4
3 4 5 6
```

SAMPLE OUTPUT

```
3 4 5
```

Question 23 Print Characters of a String

Problem Statement: Write a Java program to read a string and print each character using a "for" loop.

SAMPLE INPUT

```
Java
```

SAMPLE OUTPUT

```
J
a
v
a
```

Question 24 Count the Number of Vowels in a String

Problem Statement: Write a Java program to read a string and count the number of vowels (a, e, i, o, u) using a "for" loop.

SAMPLE INPUT

```
Programming
```

SAMPLE OUTPUT

```
3
```

Question 25 Reverse a String

Problem Statement: Write a Java program to read a string and print its reverse using a "for" loop.

SAMPLE INPUT

```
Java
```

SAMPLE OUTPUT

```
avaJ
```

`Question 26` **Count the Number of Words in a Sentence**

Problem Statement: Write a Java program to read a sentence and count the number of words using a "for" loop. Assume words are separated by a single space.

SAMPLE INPUT

```
Java is object oriented
```

SAMPLE OUTPUT

```
4
```

`Question 27` **Reverse the Words in a Sentence**

Problem Statement: Write a Java program to read a sentence and print the words in reverse order using a "for" loop.

SAMPLE INPUT

```
Java is fun
```

SAMPLE OUTPUT

```
fun is Java
```

4. BREAK AND CONTINUE STATEMENTS – Practice Questions

Question 1 — Find the First Prime Number Greater than 100

Problem Statement: Write a Java program to find and print the first prime number greater than 100. Use a loop to check successive numbers and use the "break" statement to terminate the loop immediately after the first prime number is found.

SAMPLE OUTPUT

```
101
```

Question 2 — Sum of Numbers While Skipping Multiples of 3 and 5

Problem Statement: Write a Java program to calculate the sum of all integers from 1 to 100. Use the "continue" statement to skip numbers that are divisible by both 3 and 5.

SAMPLE OUTPUT

```
5050
```

Question 3 — Login System with Three Attempts

Problem Statement: Write a Java program that prompts the user to enter a password up to 3 attempts. If correct, display "Login Successful" and break; otherwise display "Login Failed" after 3 attempts.

SAMPLE INPUT

```
java
admin
admin123
```

SAMPLE OUTPUT

```
Login Successful
```

`Question 4`

Calculate Average Until a Negative Number is Entered

Problem Statement: Continuously read integers until a negative number is entered. Use "break" and calculate the average of non-negative numbers.

SAMPLE INPUT

```
10
20
30
-1
```

SAMPLE OUTPUT

```
20.0
```

`Question 5`

Menu-Driven Application

Problem Statement: Repeatedly display menu (1. Add, 2. View, 3. Exit). Stop when user selects Exit using "break".

SAMPLE INPUT

```
1
3
```

SAMPLE OUTPUT

```
Add Selected
Application Closed
```

`Question 6`

Fibonacci Series Until a Term Exceeds 100

Problem Statement: Generate Fibonacci sequence. Stop generating numbers when a term becomes greater than 100 using "break".

SAMPLE OUTPUT

```
0 1 1 2 3 5 8 13 21 34 55 89
```

`Question 7`

Sum of Even Numbers While Skipping Multiples of 7

Problem Statement: Calculate the sum of all even numbers between 1 and 100. Use "continue" to skip multiples of 7.

SAMPLE OUTPUT

```
2156
```

`Question 8`

Count the Number of Vowels in a String

Problem Statement: Read a string and count vowels using "continue" to skip all consonants.

SAMPLE INPUT

```
Programming
```

SAMPLE OUTPUT

```
3
```

`Question 9`

Print Numbers While Skipping Multiples of 4 and 7

Problem Statement: Print numbers from 1 to 50 using "continue" to skip numbers divisible by both 4 and 7.

SAMPLE OUTPUT

```
1 2 3 4 5 ... 27 29 ... 50
```

`Question 10`
ATM Withdrawal Validation

Problem Statement: Simulate ATM withdrawal. If amount ≤ 0, display error and use "continue" to re-prompt. Otherwise process withdrawal and stop.

SAMPLE INPUT

```
-500
0
2500
```

SAMPLE OUTPUT

```
Invalid Amount
Invalid Amount
Withdrawal Successful
```

`Question 11`
Number Guessing Game with Limited Attempts

Problem Statement: Allow max 5 guesses for a secret number. If correct, print "Correct Guess" and break. Otherwise "Game Over".

SAMPLE INPUT

```
10
25
42
```

SAMPLE OUTPUT

```
Correct Guess
```

`Question 12`
Find the First Even Number Greater than 50

Problem Statement: Search for the first even number greater than 50. Once found, print it and break.

SAMPLE OUTPUT

```
52
```

5. STAR PATTERNS – Practice Questions

Question 1 Increasing Star Triangle

Problem Statement: Print a right-angled triangle of asterisks with increasing rows.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*
**
***
****
*****
```

Question 2 Decreasing Star Triangle

Problem Statement: Print an inverted right-angled triangle of asterisks.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*****
****
***
**
*
```

Question 3

Right-Aligned Star Triangle

Problem Statement: Print a right-aligned triangle of asterisks using leading spaces.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
    *
   **
  ***
 ****
*****
```

Question 4

Left-Aligned Inverted Star Triangle

Problem Statement: Print a left-aligned inverted triangle of asterisks by adding leading spaces.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*****
 ****
  ***
   **
    *
```

Question 5

Pyramid Pattern

Problem Statement: Print a centered pyramid of asterisks using nested loops.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
    *
   ***
  *****
 *******
*********
```

Question 6

Inverted Pyramid Pattern

Problem Statement: Print an inverted pyramid of asterisks using nested loops.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*********
 *******
  *****
   ***
    *
```

Question 7 Hollow Half Pyramid

Problem Statement: Print a hollow right-angled triangle of asterisks (boundary only).

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*
**
* *
*  *
*****
```

Question 8 Hollow Square Pattern

Problem Statement: Print a hollow square pattern of asterisks (border only).

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*****
*   *
*   *
*   *
*****
```

Question 9 Diamond Pattern

Problem Statement: Print a diamond pattern of asterisks for odd N.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
  *
 ***
*****
 ***
  *
```

Question 10 Hourglass Pattern

Problem Statement: Print an hourglass pattern of asterisks for odd N.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*****
 ***
  *
 ***
*****
```

6. NUMBER PATTERNS – Practice Questions

`Question 1`
Repeated Row Number Pattern

Problem Statement: Print a pattern where each row contains the row number repeated row-number times.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
1
22
333
4444
55555
```

`Question 2`
Right-Angled Number Triangle

Problem Statement: Print a right-angled triangle where each row contains numbers from 1 to the row number.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
1
12
123
1234
12345
```

Question 3

Inverted Right-Angled Number Triangle

Problem Statement: Print an inverted triangle containing numbers from 1 to N down to 1.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
12345
1234
123
12
1
```

Question 4

Floyd's Triangle

Problem Statement: Print Floyd's Triangle where numbers increase continuously starting from 1.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Question 5

Inverted Number Pyramid

Problem Statement: Print an inverted pyramid of palindromic numbers.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
123454321
1234321
12321
121
1
```

Question 6

Reverse Number Triangle

Problem Statement: Print a triangle where each row starts with the row number and counts down to 1.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
1
21
321
4321
54321
```

Question 7 Descending Number Triangle

Problem Statement: Print a triangle where each row contains numbers in descending order from N to row number.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
54321
5432
543
54
5
```

Question 8 Alternating Number Triangle

Problem Statement: Odd rows in ascending order, even rows in descending order.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
1
21
123
4321
12345
```

`Question 9` **Centered Number Pyramid**

Problem Statement: Print a centered pyramid of palindromic numbers.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
    1
   121
  12321
 1234321
123454321
```

`Question 10` **Number Diamond Pattern**

Problem Statement: Print a diamond-shaped palindromic number pattern.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
  1
 121
12321
 121
  1
```

`Question 11` **Concentric Square Number Pattern**

Problem Statement: Print a concentric square pattern where outer layers are N and decrease towards 1 in the center.

SAMPLE INPUT

```
4
```

SAMPLE OUTPUT

```
4 4 4 4 4 4 4
4 3 3 3 3 3 4
4 3 2 2 2 3 4
4 3 2 1 2 3 4
4 3 2 2 2 3 4
4 3 3 3 3 3 4
4 4 4 4 4 4 4
```

`Question 12` **Print Numbers in Spiral Order**

Problem Statement: Print elements of a 2D integer array in spiral order starting from top-left clockwise.

SAMPLE INPUT

```
3 3
1 2 3
4 5 6
7 8 9
```

SAMPLE OUTPUT

```
1 2 3 6 9 8 7 4 5
```

7. ALPHABET PATTERNS – Practice Questions

`Question 1`

Increasing Alphabet Triangle

Problem Statement: Print a right-angled triangle where each row contains the same uppercase letter repeated.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
A
BB
CCC
DDDD
EEEEE
```

`Question 2`

Decreasing Alphabet Triangle

Problem Statement: Inverted right-angled triangle starting from row N letter down to A.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
EEEEE
DDDD
CCC
BB
A
```

`Question 3` **Right-Angled Alphabet Triangle**

Problem Statement: Each row contains alphabets from 'A' up to the row letter.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
A
AB
ABC
ABCD
ABCDE
```

`Question 4` **Reverse Alphabet Triangle**

Problem Statement: Inverted alphabet triangle from A-E down to A.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
ABCDE
ABCD
ABC
AB
A
```

`Question 5` **Rectangular Alphabet Block**

Problem Statement: Print a rectangular block of sequential uppercase alphabets across rows.

SAMPLE INPUT

```
5
5
```

SAMPLE OUTPUT

```
A B C D E
F G H I J
K L M N O
P Q R S T
U V W X Y
```

`Question 6` **Shifting Diagonal Alphabet Pattern**

Problem Statement: Each row begins with the next letter and continues sequentially.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
A
BC
CDE
DEFG
EFGHI
```

`Question 7` **Hollow Alphabet Square**

Problem Statement: Print a hollow square using row letters on the border.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
A A A A A
B       B
C       C
D       D
E E E E E
```

`Question 8` **Alphabet Pyramid**

Problem Statement: Centered pyramid where each row repeats the corresponding letter.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
    A
   BBB
  CCCCC
 DDDDDDD
EEEEEEEEE
```

`Question 9` **Mirrored Alphabet Diamond**

Problem Statement: Diamond pattern of palindromic alphabets.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
  A
 ABA
ABCBA
 ABA
  A
```

`Question 10` **Letter-Digit Mixed Pattern**

Problem Statement: Odd rows contain letters, even rows contain digits equal to row number.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
A
22
CCC
4444
EEEEE
```

8. ARRAYS – Practice Questions

Question 1 Print the Elements of an Array

Problem Statement: Read elements of an integer array and print each in order.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
10 20 30 40 50
```

Question 2 Find the Length of an Array

Problem Statement: Read an integer array and determine total elements present.

SAMPLE INPUT

```
6
5 10 15 20 25 30
```

SAMPLE OUTPUT

```
6
```

Question 3 Find the Sum of All Elements in an Array

Problem Statement: Read an integer array and calculate the sum of all elements.

SAMPLE INPUT

```
5
2 4 6 8 10
```

SAMPLE OUTPUT

```
30
```

Question 4 Find the Maximum and Minimum Elements in an Array

Problem Statement: Read an integer array and determine the largest and smallest elements.

SAMPLE INPUT

```
6
45 12 78 34 9 56
```

SAMPLE OUTPUT

```
Maximum = 78
Minimum = 9
```

Question 5 Find the Average of Array Elements

Problem Statement: Read an integer array and calculate the average of all elements.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
30.0
```

Question 6 Find the Index of a Specific Element

Problem Statement: Search for a target element and print its first index, or -1 if not found.

SAMPLE INPUT

```
5
10 20 30 40 50
30
```

SAMPLE OUTPUT

```
2
```

Question 7 Count the Occurrences of an Element

Problem Statement: Count how many times a target element appears in an array.

SAMPLE INPUT

```
8
2 5 2 8 2 9 5 2
2
```

SAMPLE OUTPUT

```
4
```

Question 8 Reverse an Array

Problem Statement: Read an integer array and print elements in reverse order without modifying original.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
50 40 30 20 10
```

Question 9 Reverse an Array In-Place

Problem Statement: Modify the original array by reversing its elements and print the updated array.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
50 40 30 20 10
```

Question 10 Check Whether an Array is a Palindrome

Problem Statement: Determine whether an array reads the same left-to-right and right-to-left.

SAMPLE INPUT

```
5
1 2 3 2 1
```

SAMPLE OUTPUT

```
Palindrome
```

Question 11 Insert an Element at a Specific Position

Problem Statement: Insert an element at a valid 0-based index position and print the updated array.

SAMPLE INPUT

```
5
10 20 40 50 60
30
2
```

SAMPLE OUTPUT

```
10 20 30 40 50 60
```

Question 12 Remove an Element by Its Value

Problem Statement: Remove the first occurrence of a given element and print the resulting array.

SAMPLE INPUT

```
6
10 20 30 40 30 50
30
```

SAMPLE OUTPUT

```
10 20 40 30 50
```

Question 13 Remove All Occurrences of a Specific Element

Problem Statement: Remove all occurrences of a specified element and print the updated array.

SAMPLE INPUT

```
7
2 4 2 6 2 8 2
2
```

SAMPLE OUTPUT

```
4 6 8
```

`Question 14` **Add an Element to the End of an Array**

Problem Statement: Append a new element to the end of the array and print the updated array.

SAMPLE INPUT

```
4
5 10 15 20
25
```

SAMPLE OUTPUT

```
5 10 15 20 25
```

`Question 15` **Merge Two Arrays**

Problem Statement: Merge two integer arrays into a single array preserving element order.

SAMPLE INPUT

```
3
10 20 30
4
40 50 60 70
```

SAMPLE OUTPUT

```
10 20 30 40 50 60 70
```

Question 16 — Remove Elements at Even Indices

Problem Statement: Create a new array removing all elements at even indices (0, 2, 4...).

SAMPLE INPUT

```
6
10 20 30 40 50 60
```

SAMPLE OUTPUT

```
20 40 60
```

Question 17 — Slice an Array

Problem Statement: Create a sub-array using Arrays.copyOfRange() with start and end indices.

SAMPLE INPUT

```
7
10 20 30 40 50 60 70
2 5
```

SAMPLE OUTPUT

```
30 40 50
```

Question 18 — Sort an Array in Ascending Order

Problem Statement: Sort array elements in ascending order using Arrays.sort().

SAMPLE INPUT

```
6
45 12 67 8 34 23
```

SAMPLE OUTPUT

```
8 12 23 34 45 67
```

Question 19 — Sort an Array in Descending Order

Problem Statement: Sort array elements in descending order.

SAMPLE INPUT

```
6
45 12 67 8 34 23
```

SAMPLE OUTPUT

```
67 45 34 23 12 8
```

Question 20 — Find the Second Largest Element

Problem Statement: Find the second largest distinct element in an array.

SAMPLE INPUT

```
6
12 45 67 23 89 56
```

SAMPLE OUTPUT

```
67
```

Question 21 — Find the Missing Number

Problem Statement: Find the single missing number from an array of numbers from 1 to N.

SAMPLE INPUT

```
6
1 2 3 5 6
```

SAMPLE OUTPUT

```
4
```

`Question 22` **Check Whether Two Arrays are Equal**

Problem Statement: Compare two integer arrays of same size using Arrays.equals().

SAMPLE INPUT

```
5
10 20 30 40 50
10 20 30 40 50
```

SAMPLE OUTPUT

```
Equal
```

`Question 23` **Find the Intersection of Two Arrays**

Problem Statement: Print common elements present in both arrays.

SAMPLE INPUT

```
5
1 2 3 4 5
5
3 4 5 6 7
```

SAMPLE OUTPUT

```
3 4 5
```

`Question 24` **Find the Union of Two Arrays**

Problem Statement: Print all distinct elements present across two arrays.

SAMPLE INPUT

```
5
1 2 3 4 5
5
4 5 6 7 8
```

SAMPLE OUTPUT

```
1 2 3 4 5 6 7 8
```

Question 25 Find the Frequency of Each Element

Problem Statement: Determine frequency of each distinct element using a HashMap.

SAMPLE INPUT

```
8
2 3 2 5 3 2 5 4
```

SAMPLE OUTPUT

```
2 : 3
3 : 2
5 : 2
4 : 1
```

9. STRINGS – Basic Operations Practice Questions

Question 1 Display a String

Read a string from user and display it.

SAMPLE INPUT

```
Hello Java
```

SAMPLE OUTPUT

```
Hello Java
```

Question 2 Find the Length of a String

Determine total characters present using length().

SAMPLE INPUT

```
Programming
```

SAMPLE OUTPUT

```
11
```

Question 3 Print First and Last Character

Print first and last character using charAt().

SAMPLE INPUT

```
Computer
```

SAMPLE OUTPUT

```
First Character: C
Last Character: r
```

Question 4 Reverse a String

Print string characters in reverse order.

SAMPLE INPUT

```
Java
```

SAMPLE OUTPUT

```
avaJ
```

`Question 5`

Check Whether a String is a Palindrome

Determine whether string reads same left-to-right and right-to-left.

SAMPLE INPUT

```
madam
```

SAMPLE OUTPUT

```
Palindrome
```

`Question 6`

Check Palindrome Ignoring Spaces and Case

Check palindrome after removing spaces and converting to lowercase.

SAMPLE INPUT

```
Never Odd Or Even
```

SAMPLE OUTPUT

```
Palindrome
```

`Question 7`

Convert String to Uppercase

Convert lowercase letters to uppercase using toUpperCase().

SAMPLE INPUT

```
Java Programming
```

SAMPLE OUTPUT

```
JAVA PROGRAMMING
```

Question 8 Convert String to Lowercase

Convert uppercase letters to lowercase using toLowerCase().

SAMPLE INPUT

```
JAVA Programming
```

SAMPLE OUTPUT

```
java programming
```

Question 9 Trim Leading and Trailing Spaces

Remove unwanted leading and trailing whitespace using trim().

SAMPLE INPUT

```
   Welcome to Java
```

SAMPLE OUTPUT

```
Welcome to Java
```

Question 10 Check Whether String is Empty or Blank

Check if string is empty or blank using isEmpty() / isBlank().

SAMPLE OUTPUT

```
Blank String
```

Question 11

Check Whether Character Exists

Check if given character exists in the string.

SAMPLE INPUT

```
Programming
g
```

SAMPLE OUTPUT

```
Character Found
```

Question 12

Count Occurrences of a Character

Count occurrences of a specific character in string.

SAMPLE INPUT

```
Mississippi
s
```

SAMPLE OUTPUT

```
4
```

Question 13

Count Vowels in a String

Count total vowels present in string (both upper/lowercase).

SAMPLE INPUT

```
OpenAI ChatGPT
```

SAMPLE OUTPUT

```
5
```

Question 14

Count Consonants in a String

Count total consonants in string ignoring digits and spaces.

SAMPLE INPUT

```
Java Programming
```

SAMPLE OUTPUT

```
10
```

Question 15

Remove All Vowels from a String

Remove all vowels preserving remaining characters.

SAMPLE INPUT

```
Programming
```

SAMPLE OUTPUT

```
Prgrmmng
```

Question 16

Count Number of Words in a Sentence

Determine total words present in sentence.

SAMPLE INPUT

```
Java is easy to learn
```

SAMPLE OUTPUT

```
5
```

`Question 17` **Find the Longest Word in a Sentence**

Identify longest word in sentence (first occurrence if tie).

SAMPLE INPUT

```
Java programming language
```

SAMPLE OUTPUT

```
programming
```

`Question 18` **Reverse Order of Words in a Sentence**

Print words in reverse order without reversing individual words.

SAMPLE INPUT

```
Java is fun
```

SAMPLE OUTPUT

```
fun is Java
```

10. STRINGS – String Manipulation Practice Questions

`Question 19`

Concatenate Two Strings

Concatenate using + operator and concat() method.

SAMPLE INPUT

```
Hello
World
```

SAMPLE OUTPUT

```
HelloWorld
```

`Question 20`

Repeat a String N Times

Repeat string N times using repeat().

SAMPLE INPUT

```
Hi
3
```

SAMPLE OUTPUT

```
HiHiHi
```

`Question 21`

Replace Character or Substring

Replace occurrences using replace() method.

SAMPLE INPUT

```
Java Programming
Java
Python
```

SAMPLE OUTPUT

```
Python Programming
```

Question 22 Split Sentence into Words

Split sentence using split() and print each word on new line.

SAMPLE INPUT

```
Java is platform independent
```

SAMPLE OUTPUT

```
Java
is
platform
independent
```

Question 23 Extract Substring

Extract substring using start and end indices with substring().

SAMPLE INPUT

```
Programming
3
8
```

SAMPLE OUTPUT

```
gramm
```

Question 24 Convert String to Character Array

Convert to char array using toCharArray() and print each character.

SAMPLE INPUT

```
Java
```

SAMPLE OUTPUT

```
J
a
v
a
```

Question 25 Count Occurrence of Each Character

Determine frequency of each distinct character in string.

SAMPLE INPUT

```
banana
```

SAMPLE OUTPUT

```
b : 1
a : 3
n : 2
```

Question 26 Find All Occurrences of a Substring

Find all starting indices of substring in string.

SAMPLE INPUT

```
bananaban
ban
```

SAMPLE OUTPUT

```
0
6
```

Question 27 Extract All Numbers from a String

Extract all numeric values from alphanumeric string.

SAMPLE INPUT

```
Student25 scored98 in2025
```

SAMPLE OUTPUT

```
25 98 2025
```

Question 28

Determine Whether Two Strings are Anagrams

Check if two strings contain same characters with same frequencies.

SAMPLE INPUT

```
Listen
Silent
```

SAMPLE OUTPUT

```
Anagrams
```

Question 29

Encode String Using Caesar Cipher (Shift 1)

Shift each alphabet by 1 ('Z'→'A', 'z'→'a').

SAMPLE INPUT

```
Hello Zebra
```

SAMPLE OUTPUT

```
Ifmmp Afcsb
```

Question 30

Compress String Using Character Counts

Replace consecutive repeated characters with char + count.

SAMPLE INPUT

```
aabcccccaaa
```

SAMPLE OUTPUT

```
a2b1c5a3
```

11. STRINGBUILDER – Practice Questions

`Question 1`

Append Multiple Values

Append string, integer, character using append().

SAMPLE INPUT

```
Java
2025
!
```

SAMPLE OUTPUT

```
Java2025!
```

`Question 2`

Insert a String at a Specific Position

Insert string at index using insert().

SAMPLE INPUT

```
HelloWorld
Java
5
```

SAMPLE OUTPUT

```
HelloJavaWorld
```

`Question 3`

Delete a Range of Characters

Delete specified character range using delete().

SAMPLE INPUT

```
Programming
3
7
```

SAMPLE OUTPUT

```
Proming
```

`Question 4`

Reverse a StringBuilder

Reverse string stored in StringBuilder using reverse().

SAMPLE INPUT

```
Java
```

SAMPLE OUTPUT

```
avaJ
```

`Question 5`

Replace Part of a String

Replace range with new string using replace().

SAMPLE INPUT

```
HelloWorld
5
10
Java
```

SAMPLE OUTPUT

```
HelloJava
```

`Question 6`

Convert StringBuilder to String

Convert StringBuilder to String using toString().

SAMPLE INPUT

```
Java Programming
```

SAMPLE OUTPUT

```
Java Programming
```

`Question 7`

Compare StringBuilder and String Concatenation

Compare execution time of repeated N concatenations between String and StringBuilder.

SAMPLE INPUT

```
10000
```

SAMPLE OUTPUT

```
String Time: 45 ms
StringBuilder Time: 2 ms
```

`Question 8`

Build a CSV Line

Construct comma-separated CSV line using StringBuilder.

SAMPLE INPUT

```
4
John
25
Hyderabad
India
```

SAMPLE OUTPUT

```
John,25,Hyderabad,India
```

`Question 9`

Modify a Character Using setCharAt()

Modify character at specific index using setCharAt().

SAMPLE INPUT

```
Java
1
o
```

SAMPLE OUTPUT

```
Jova
```

`Question 10` **Build a Large Repeated Pattern**

Construct string by repeating pattern N times using StringBuilder.

SAMPLE INPUT

```
AB
5
```

SAMPLE OUTPUT

```
ABABABABAB
```

12. CLASSES AND OBJECTS – Practice Questions

`Question 1` **Create a Simple Class with Fields, Constructor, and Methods**

Create a Student class with name and age fields, constructor, and display method.

SAMPLE INPUT

```
Rahul
20
```

SAMPLE OUTPUT

```
Student Name: Rahul
Age: 20
```

Question 2 Create an Object and Access Its Fields and Methods

Create an Employee class with employee ID and name. Assign values and display.

SAMPLE INPUT

```
101
Anita
```

SAMPLE OUTPUT

```
Employee ID: 101
Employee Name: Anita
```

Question 3 Create a Person Class

Create Person class with name and age. Read from user and display details.

SAMPLE INPUT

```
Kiran
25
```

SAMPLE OUTPUT

```
Name: Kiran
Age: 25
```

Question 4 Create a Student Class with Average Calculation

Student class with name, roll number, and 3 subject marks. Calculate average.

SAMPLE INPUT

```
Sneha
25
80
90
85
```

SAMPLE OUTPUT

```
Student Name: Sneha
Roll Number: 25
Average Marks: 85.0
```

Question 5 Create an Employee Class with Salary Increment

Employee class with ID, name, salary. Method to increase salary by given amount.

SAMPLE INPUT

```
102
Ravi
35000
5000
```

SAMPLE OUTPUT

```
Employee ID: 102
Employee Name: Ravi
Updated Salary: 40000.0
```

Question 6 Create a BankAccount Class

BankAccount class with deposit(), withdraw(), and current balance methods.

SAMPLE INPUT

```
1001
Arun
5000
2000
1500
```

SAMPLE OUTPUT

```
Current Balance: 5500.0
```

Question 7 Create a Car Class

Car class with make, model, speed. Implement accelerate() and brake().

SAMPLE INPUT

```
Toyota
Innova
40
30
10
```

SAMPLE OUTPUT

```
Current Speed: 60 km/h
```

Question 8 Create a Book Class

Book class with title, author, year. Method to display details.

SAMPLE INPUT

```
Java Programming
James Gosling
2022
```

SAMPLE OUTPUT

```
Title: Java Programming
Author: James Gosling
Publication Year: 2022
```

Question 9 Create a LibraryBook Class

LibraryBook class with availability status, issue(), and returnBook().

SAMPLE INPUT

```
Data Structures
Mark Allen
Issue
```

SAMPLE OUTPUT

```
Book Issued
Availability: Not Available
```

Question 10 Create a Shop Class

Shop class storing multiple product names. Add and display all products.

SAMPLE INPUT

```
3
Laptop
Mouse
Keyboard
```

SAMPLE OUTPUT

```
Available Products:
Laptop
Mouse
Keyboard
```

Question 11 Create a Calculator Class

Calculator class with addition, subtraction, multiplication, and division methods.

SAMPLE INPUT

```
20
5
```

SAMPLE OUTPUT

```
Addition: 25
Subtraction: 15
Multiplication: 100
Division: 4.0
```

Question 12 Create a Rectangle Class

Rectangle class with length, breadth. Calculate area and perimeter.

SAMPLE INPUT

```
8
5
```

SAMPLE OUTPUT

```
Area: 40.0
Perimeter: 26.0
```

`Question 13` **Create a Circle Class**

Circle class with radius. Calculate area and circumference ($\pi=3.14$).

SAMPLE INPUT

```
7
```

SAMPLE OUTPUT

```
Area: 153.86
Circumference: 43.96
```

`Question 14` **Create a TemperatureConverter Class**

TemperatureConverter class with Celsius to Fahrenheit and vice versa methods.

SAMPLE INPUT

```
37
C
```

SAMPLE OUTPUT

```
98.6
```

`Question 15` **Create a Vector2D Class**

Vector2D class with x, y coordinates. Calculate magnitude = $\sqrt{(x^2 + y^2)}$.

SAMPLE INPUT

```
3
4
```

SAMPLE OUTPUT

```
Magnitude: 5.0
```

13. METHODS (FUNCTIONS) – Practice Questions

I. Methods with No Parameters and No Return Value

Question 1 Display a Welcome Message

Create void displayMessage() that prints "Welcome to Java Programming".

SAMPLE OUTPUT

```
Welcome to Java Programming
```

Question 2 Print Multiplication Table of 5

Create void printTable() that prints table of 5 from 1 to 10.

SAMPLE OUTPUT

```
5 x 1 = 5 ... 5 x 10 = 50
```

Question 3 Print Even Numbers from 1 to 100

Create void printEvenNumbers() that prints even numbers from 1 to 100.

SAMPLE OUTPUT

```
2 4 6 8 ... 100
```

Question 4

Print the Fibonacci Series

Create void printFibonacci() that reads N and prints first N Fibonacci numbers.

SAMPLE INPUT

```
7
```

SAMPLE OUTPUT

```
0 1 1 2 3 5 8
```

Question 5

Display Current Date and Time

Create void displayDateTime() that displays current system date and time.

SAMPLE OUTPUT

```
2026-07-23 10:30:45
```

Question 6

Print a Right-Angled Star Pattern

Create void printPattern() that reads row count and prints star pattern.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
*
**
***
****
*****
```

`Question 7` **Print First N Prime Numbers**

Create void printPrimeNumbers() that reads N and prints first N prime numbers.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
2 3 5 7 11
```

`Question 8` **Display Student Information**

Create void displayStudent() that stores name, roll number, department and prints.

SAMPLE OUTPUT

```
Name: Rahul
Roll Number: 101
Department: CSE
```

II. Methods with Parameters and No Return Value

`Question 9` **Print Multiplication Table of Given Number**

void printTable(int number) prints multiplication table up to 10.

SAMPLE INPUT

```
7
```

SAMPLE OUTPUT

```
7 x 1 = 7 ... 7 x 10 = 70
```

Question 10 Display Sum of Two Numbers

void displaySum(int a, int b) prints sum of two integers.

SAMPLE INPUT

```
15
25
```

SAMPLE OUTPUT

```
Sum = 40
```

Question 11 Display Larger of Two Numbers

void displayLargest(int a, int b) displays larger integer.

SAMPLE INPUT

```
48
72
```

SAMPLE OUTPUT

```
Largest Number = 72
```

Question 12 Check Whether Number is Even or Odd

void checkEvenOdd(int number) displays Even or Odd.

SAMPLE INPUT

```
39
```

SAMPLE OUTPUT

```
Odd
```

Question 13 Print a Rectangle Pattern

void printRectangle(int rows, int columns) prints asterisk rectangle.

SAMPLE INPUT

```
4
6
```

SAMPLE OUTPUT

```
******
******
******
******
```

Question 14 Display Factorial of a Number

void displayFactorial(int number) calculates and displays factorial.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
Factorial = 120
```

Question 15 Display Prime Numbers Within Range

void displayPrimes(int start, int end) displays all prime numbers in range.

SAMPLE INPUT

```
10
30
```

SAMPLE OUTPUT

```
11 13 17 19 23 29
```

Question 16 Display Reverse of a Number

void reverseNumber(int number) displays reversed integer.

SAMPLE INPUT

```
54821
```

SAMPLE OUTPUT

```
12845
```

Question 17 Print Elements of Integer Array

void printArray(int[] array) displays all array elements.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
10 20 30 40 50
```

Question 18 Display Any Number of Strings Using Varargs

void displayStrings(String... words) displays each string on a new line.

SAMPLE INPUT

```
4
Java
Python
C++
Go
```

SAMPLE OUTPUT

```
Java
Python
C++
Go
```

III. Methods with No Parameters and Return Value

`Question 19` **Return a Random Integer**

int getRandomNumber() returns a random integer between 1 and 100.

SAMPLE OUTPUT

```
57
```

`Question 20` **Return Current Year**

int getCurrentYear() returns current system year.

SAMPLE OUTPUT

```
2026
```

`Question 21` **Read and Return a Number**

int readNumber() reads an integer from user and returns it.

SAMPLE INPUT

```
45
```

SAMPLE OUTPUT

```
45
```

Question 22 Return Factorial of a Number

long getFactorial() reads non-negative integer inside method and returns factorial.

SAMPLE INPUT

```
6
```

SAMPLE OUTPUT

```
720
```

Question 23 Return Largest Element in an Array

int getLargestElement() reads array inside method and returns largest element.

SAMPLE INPUT

```
6
12 45 8 67 23 40
```

SAMPLE OUTPUT

```
67
```

Question 24 Return Reverse of a String

String reverseString() reads string inside method and returns reversed string.

SAMPLE INPUT

```
Programming
```

SAMPLE OUTPUT

```
gnimmargorP
```

Question 25 Return Average of Marks

double getAverage() reads 3 subject marks inside method and returns average.

SAMPLE INPUT

```
80
90
85
```

SAMPLE OUTPUT

```
85.0
```

Question 26 Return Whether a String is a Palindrome

boolean isPalindrome() reads string inside method and returns true/false.

SAMPLE INPUT

```
madam
```

SAMPLE OUTPUT

```
Palindrome
```

IV. Methods with Parameters and Return Value

Question 27 Return Sum of Two Numbers

int add(int a, int b) returns sum of two integers.

SAMPLE INPUT

```
25
35
```

SAMPLE OUTPUT

```
60
```

Question 28 Return Larger of Two Numbers

int findLargest(int a, int b) returns larger value.

SAMPLE INPUT

```
84
67
```

SAMPLE OUTPUT

```
84
```

Question 29 Return Factorial of a Number

long factorial(int number) returns factorial of non-negative integer.

SAMPLE INPUT

```
5
```

SAMPLE OUTPUT

```
120
```

Question 30 Return nth Fibonacci Number Using Recursion

int fibonacci(int n) recursive method returning nth Fibonacci number.

SAMPLE INPUT

```
8
```

SAMPLE OUTPUT

```
13
```

Question 31 — Check Whether a Number is Prime

boolean isPrime(int number) returns true if prime, false otherwise.

SAMPLE INPUT

```
29
```

SAMPLE OUTPUT

```
Prime
```

Question 32 — Return Sum of Digits Using Recursion

int sumOfDigits(int number) recursive method returning sum of digits.

SAMPLE INPUT

```
5678
```

SAMPLE OUTPUT

```
26
```

Question 33 — Return Sum of Array Elements

int sumArray(int[] array) returns sum of array elements.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
150
```

Question 34 Return Maximum Value Using Varargs

int findMaximum(int... numbers) returns largest value using varargs.

SAMPLE INPUT

```
6
15 42 8 91 37 25
```

SAMPLE OUTPUT

```
91
```

Question 35 Return Average Using Varargs

double calculateAverage(double... values) returns average of numeric values.

SAMPLE INPUT

```
4
85 90 80 95
```

SAMPLE OUTPUT

```
87.5
```

Question 36 Overload Methods to Calculate Area

Overload area() for square (1 int), rectangle (2 ints), and circle (1 double, π=3.14).

SAMPLE INPUT

```
5
8 4
7
```

SAMPLE OUTPUT

```
Square Area: 25
Rectangle Area: 32
Circle Area: 153.86
```

`Question 37` **Count Number of Digits Using Recursion**

int countDigits(int number) recursive method returning total digit count.

SAMPLE INPUT	SAMPLE OUTPUT
`987654`	`6`

`Question 38` **Check Palindrome String Using Recursion**

boolean isPalindrome(String text, int left, int right) recursive palindrome check.

SAMPLE INPUT	SAMPLE OUTPUT
`level`	`Palindrome`

14. STATIC METHODS – Practice Questions

Question 1 Create a Calculator Class Using Static Methods

Calculator class with static methods for add, subtract, multiply, divide. Call via class name.

SAMPLE INPUT

```
20
5
```

SAMPLE OUTPUT

```
Addition: 25
Subtraction: 15
Multiplication: 100
Division: 4.0
```

Question 2 Create a NumberUtility Class

NumberUtility class with static method checking even or odd.

SAMPLE INPUT

```
37
```

SAMPLE OUTPUT

```
Odd
```

Question 3 Create a TemperatureConverter Class

TemperatureConverter with two static methods converting C→F and F→C.

SAMPLE INPUT

```
25
C
```

SAMPLE OUTPUT

```
77.0
```

`Question 4`

Create a StringAnalyzer Class

StringAnalyzer static method checking palindrome ignoring spaces/case/punctuation.

SAMPLE INPUT

```
A man, a plan, a canal: Panama
```

SAMPLE OUTPUT

```
Palindrome
```

`Question 5`

Create a Fibonacci Class

Fibonacci static method printing first N Fibonacci terms.

SAMPLE INPUT

```
7
```

SAMPLE OUTPUT

```
0 1 1 2 3 5 8
```

`Question 6`

Create a PrimeUtility Class

PrimeUtility with static isPrime() and static getNthPrime() methods.

SAMPLE INPUT

```
29
10
```

SAMPLE OUTPUT

```
Prime
29
```

`Question 7`

Create a Geometry Class

Geometry class with static area methods for circle (π=3.14), rectangle, triangle.

SAMPLE INPUT

```
7
8 5
10 6
```

SAMPLE OUTPUT

```
Circle Area: 153.86
Rectangle Area: 40.0
Triangle Area: 30.0
```

`Question 8`

Create a DateValidator Class

DateValidator static method validating "YYYY-MM-DD" format date.

SAMPLE INPUT

```
2026-07-23
```

SAMPLE OUTPUT

```
Valid Date
```

`Question 9`

Create a DateOperations Class

DateOperations static method calculating days between two dates (assume 30 days/month).

SAMPLE INPUT

```
2026-07-01
2026-07-20
```

SAMPLE OUTPUT

```
19
```

`Question 10` **Create a BaseConverter Class**

BaseConverter static method converting number from source base to destination base (2–16).

SAMPLE INPUT

```
101101
2
10
```

SAMPLE OUTPUT

```
45
```

15. ENCAPSULATION, GETTERS, SETTERS & VALIDATION – Practice Questions

`Question 1` **Create a Student Class with Validated Age**

Private name/age fields. Setter ensures age is between 5 and 25 inclusive.

SAMPLE INPUT

```
Rahul
20
```

SAMPLE OUTPUT

```
Name: Rahul
Age: 20
```

Question 2 Create a BankAccount Class with Controlled Balance

Private balance with getter only. deposit() and withdraw() methods rejecting negative balance.

SAMPLE INPUT

```
5000
2000
1500
```

SAMPLE OUTPUT

```
Current Balance: 5500.0
```

Question 3 Create a Circle Class with Radius Validation

Private radius. Setter rejects negative radius. Area method (π=3.14).

SAMPLE INPUT

```
7
```

SAMPLE OUTPUT

```
Area: 153.86
```

Question 4 Create a Laptop Class with Battery Management

Private batteryPercentage. charge() and use() keeping level between 0 and 100.

SAMPLE INPUT

```
60
30
20
```

SAMPLE OUTPUT

```
Battery Percentage: 70%
```

Question 5

Create UserProfile Class with Password Validation

Private username, password. Password setter requires at least 8 characters.

SAMPLE INPUT

```
admin
Password123
```

SAMPLE OUTPUT

```
User Profile Created Successfully
```

Question 6

Employee Class with Salary and Bonus Validation

Private salary, bonus. Bonus setter ensures bonus does not exceed 20% of salary.

SAMPLE INPUT

```
50000
8000
```

SAMPLE OUTPUT

```
Salary: 50000.0
Bonus: 8000.0
```

Question 7

Create Temperature Class with Unit Conversion

Store internally in Celsius. Provide setFahrenheit() and getFahrenheit() conversions.

SAMPLE INPUT

```
98.6
```

SAMPLE OUTPUT

```
Celsius: 37.0
Fahrenheit: 98.6
```

Question 8 — Create Square Class with Derived Properties

Private side. Implement getArea() and getPerimeter() dynamically.

SAMPLE INPUT

```
6
```

SAMPLE OUTPUT

```
Area: 36.0
Perimeter: 24.0
```

Question 9 — Smartphone Class to Track Storage Usage

Private totalStorage, storageUsed. Getters compute available storage and percentage used.

SAMPLE INPUT

```
128
50
```

SAMPLE OUTPUT

```
Available Storage: 78 GB
Storage Used: 39.06%
```

Question 10 — Library Class for Book Management

Private lists for available and issued books. Methods to issue, return, and status.

SAMPLE INPUT

```
3
Java
Python
C++
Java
Java
```

SAMPLE OUTPUT

```
Available Books:
Python
C++
Issued Books:
Java
```

16. INHERITANCE – Practice Questions

Question 1 — Create Single Inheritance hierarchy with Vehicle and Car

Vehicle parent class with brand/speed. Car child class adds doors count.

SAMPLE INPUT

```
Honda
120
4
```

SAMPLE OUTPUT

```
Brand: Honda
Speed: 120 km/h
Doors: 4
```

Question 2 — Create Multilevel Inheritance with Person, Employee, Manager

Person (name) → Employee (salary) → Manager (department).

SAMPLE INPUT

```
Vikram
75000
IT
```

SAMPLE OUTPUT

```
Name: Vikram
Salary: 75000.0
Department: IT
```

Question 3 — Create Hierarchical Inheritance with Shape, Circle, Rectangle

Shape parent. Circle (radius) and Rectangle (length, breadth) children.

SAMPLE INPUT

```
Circle
7
Rectangle
5 8
```

SAMPLE OUTPUT

```
Circle Area: 153.86
Rectangle Area: 40.0
```

Question 4 — Demonstrate Use of super Keyword in Constructors

Use super() to invoke parent constructor and super.method() to call parent method.

SAMPLE INPUT

```
Animal
Dog
Bark
```

SAMPLE OUTPUT

```
Type: Animal
Name: Dog
Sound: Bark
```

Question 5 — BankAccount and SavingsAccount

BankAccount parent. SavingsAccount child adds interestRate and calculateInterest().

SAMPLE INPUT

```
10000
5.0
```

SAMPLE OUTPUT

```
Annual Interest: 500.0
```

Question 6 Appliance and WashingMachine

Appliance (brand, power) → WashingMachine (capacityInKg).

SAMPLE INPUT

```
LG
1500
7
```

SAMPLE OUTPUT

```
Brand: LG
Power Consumption: 1500 W
Capacity: 7 kg
```

Question 7 Book and EBook

Book (title, author) → EBook (fileSizeInMB, downloadFormat).

SAMPLE INPUT

```
Clean Code
Robert Martin
15
PDF
```

SAMPLE OUTPUT

```
Title: Clean Code
Author: Robert Martin
File Size: 15 MB
Format: PDF
```

Question 8 Building and ResidentialBuilding

Building (address, floors) → ResidentialBuilding (apartmentsCount).

SAMPLE INPUT

```
MG Road, Bangalore
10
40
```

SAMPLE OUTPUT

```
Address: MG Road, Bangalore
Floors: 10
Total Apartments: 40
```

Question 9 Food and FastFood

Food (name, calories) → FastFood (isJunkFood boolean).

SAMPLE INPUT

```
Burger
450
true
```

SAMPLE OUTPUT

```
Food: Burger
Calories: 450
Type: Fast Food
```

Question 10 Account, SavingsAccount, FixedDepositAccount

Account parent → SavingsAccount & FixedDepositAccount children with interest calculations.

SAMPLE INPUT

```
Savings
50000
4.0
FD
100000
7.5
2
```

SAMPLE OUTPUT

```
Savings Interest: 2000.0
FD Interest: 15000.0
```

17. ABSTRACTION – Practice Questions

Question 1

Abstract Class Shape with Subclasses Circle and Rectangle

Abstract method calculateArea(). Subclasses implement formula.

SAMPLE INPUT

```
Circle
5
Rectangle
4 6
```

SAMPLE OUTPUT

```
Circle Area: 78.5
Rectangle Area: 24.0
```

Question 2

Interface Playable Implemented by Guitar and Piano

Interface Playable with play() method. Guitar and Piano implement custom sounds.

SAMPLE OUTPUT

```
Guitar plays strum sound
Piano plays key sound
```

Question 3 Abstract Class BankAccount with SavingsAccount and CurrentAccount

Abstract calculateInterest(). Savings has fixed %, Current has no interest.

SAMPLE INPUT

```
Savings
20000
5
Current
50000
```

SAMPLE OUTPUT

```
Savings Interest: 1000.0
Current Interest: 0.0
```

Question 4 Interfaces Resizable and Rotatable Implemented by Image

Demonstrate multiple interface implementation on Image class.

SAMPLE INPUT

```
50
90
```

SAMPLE OUTPUT

```
Resized to 50%
Rotated by 90 degrees
```

Question 5 Abstract Class Employee with FullTimeEmployee and PartTimeEmployee

Abstract calculateSalary(). FullTime = monthly salary, PartTime = hours * rate.

SAMPLE INPUT

```
FullTime
60000
PartTime
80
500
```

SAMPLE OUTPUT

```
Full-Time Salary: 60000.0
Part-Time Salary: 40000.0
```

Question 6 Interface Printable Implemented by Document and Image

Printable interface with print() method.

SAMPLE OUTPUT

```
Printing Document content...
Printing Image file...
```

`Question 7`

Interface PaymentGateway Implemented by CreditCard and UPI

PaymentGateway with processPayment(amount).

SAMPLE INPUT

```
CreditCard
2500
UPI
1000
```

SAMPLE OUTPUT

```
Paid 2500.0 using Credit Card
Paid 1000.0 using UPI
```

`Question 8`

Abstract Class Appliance with WashingMachine and Microwave

Abstract turnOn() and turnOff() methods.

SAMPLE OUTPUT

```
Washing Machine started washing
Microwave started heating
```

`Question 9`

Interface Sortable Implemented by ArraySorter

Sortable with sort(int[] arr) method.

SAMPLE INPUT

```
5
4 2 8 1 5
```

SAMPLE OUTPUT

```
1 2 4 5 8
```

Question 10

Interface DatabaseConnection for MySQL and PostgreSQL

Interface DatabaseConnection with connect() and disconnect() methods.

SAMPLE OUTPUT

```
Connected to MySQL Database
Connected to PostgreSQL Database
```

18. POLYMORPHISM – Practice Questions

Question 1

Method Overloading with Math Operations

Compile-time polymorphism: multiply() overloaded for 2 ints, 3 ints, 2 doubles.

SAMPLE INPUT

```
4 5
2 3 4
2.5 4.0
```

SAMPLE OUTPUT

```
Result 1: 20
Result 2: 24
Result 3: 10.0
```

Question 2 Method Overriding with Animal Hierarchy

Runtime polymorphism: Animal reference calling makeSound() on Dog and Cat objects.

SAMPLE OUTPUT

```
Dog barks
Cat meows
```

Question 3 Dynamic Method Dispatch with Shape Reference

Shape array storing Circle, Square, Triangle calling draw().

SAMPLE OUTPUT

```
Drawing Circle
Drawing Square
Drawing Triangle
```

Question 4 Overloaded Area Method for Geometric Shapes

Overload calculateArea() for Circle, Rectangle, and Triangle.

SAMPLE INPUT

```
7
4 6
5 8
```

SAMPLE OUTPUT

```
Circle Area: 153.86
Rectangle Area: 24.0
Triangle Area: 20.0
```

Question 5 Polymorphic Payment System

Payment class parent, CreditCard, DebitCard, NetBanking children override pay().

SAMPLE INPUT

```
CreditCard
1500
```

SAMPLE OUTPUT

```
Paid 1500.0 via Credit Card
```

Question 6 Overloaded Search Methods in LibrarySystem

search() overloaded by title (String), author (String, String), publication year (int).

SAMPLE INPUT

```
Java Programming
```

SAMPLE OUTPUT

```
Found Book: Java Programming
```

Question 7 Polymorphic Notification System

Notification parent. EmailNotification, SMSNotification override send().

SAMPLE INPUT

```
Hello User!
```

SAMPLE OUTPUT

```
Sending Email: Hello User!
Sending SMS: Hello User!
```

Question 8 Polymorphic Order Processing

Order parent. OnlineOrder and InStoreOrder override processOrder().

SAMPLE OUTPUT

```
Processing Online Order: Home Deliv
ery Scheduled
Processing In-Store Order: Receipt
Generated
```

Question 9 Overloaded Constructor in Student Class

Constructors with default, name-only, and name+age+roll parameters.

SAMPLE OUTPUT

```
Student 1: Unknown, 0, 0
Student 2: Anita, 0, 0
Student 3: Rahul, 20, 101
```

Question 10 Polymorphic Vehicle Rental System

Vehicle parent. Car and Bike override calculateRent(int days).

SAMPLE INPUT

```
Car
3
Bike
5
```

SAMPLE OUTPUT

```
Car Rent: 4500.0
Bike Rent: 2500.0
```

19. METHOD OVERRIDING – Practice Questions

Question 1 Override toString() Method

Override toString() in Person class to print formatted details.

SAMPLE INPUT

```
Rahul
20
```

SAMPLE OUTPUT

```
Person[Name=Rahul, Age=20]
```

Question 2 Override equals() Method

Override equals() in Student class comparing roll number and department.

SAMPLE INPUT

```
101 CSE
101 CSE
```

SAMPLE OUTPUT

```
Objects are Equal
```

Question 3 Override hashCode() along with equals()

Ensure equal objects produce identical hash codes for Employee class.

SAMPLE INPUT

```
101
101
```

SAMPLE OUTPUT

```
HashCodes are Equal
```

Question 4 Bank and Interest Rates

Bank parent (getInterestRate=0%). SBI, ICICI, HDFC override with specific rates.

SAMPLE OUTPUT

```
SBI Interest Rate: 6.5%
ICICI Interest Rate: 7.0%
HDFC Interest Rate: 6.8%
```

Question 5 Vehicle and Speed Limits

Vehicle maxSpeed() overridden by Car (180), SportsCar (320), Truck (100).

SAMPLE OUTPUT

```
Car Max Speed: 180 km/h
SportsCar Max Speed: 320 km/h
Truck Max Speed: 100 km/h
```

Question 6 Employee and Manager Bonus Calculations

Employee calculateBonus() = 10% salary. Manager overrides to 20% + fixed stipend.

SAMPLE INPUT

```
50000
80000
```

SAMPLE OUTPUT

```
Employee Bonus: 5000.0
Manager Bonus: 21000.0
```

Question 7

Shape and Draw Operations

Shape draw() overridden by Circle, Square, and Polygon.

SAMPLE OUTPUT

```
Drawing Circle with radius
Drawing Square with side
Drawing Polygon with sides
```

Question 8

Account and Tax Deduction

Account calculateTax() overridden by SavingsAccount (0%) and DematAccount (15%).

SAMPLE INPUT

```
50000
```

SAMPLE OUTPUT

```
Savings Tax: 0.0
Demat Tax: 7500.0
```

Question 9

Printer and Specialized Printers

Printer print() overridden by LaserPrinter and InkjetPrinter.

SAMPLE OUTPUT

```
LaserPrinter printing fast monochro
me
InkjetPrinter printing high quality
color
```

`Question 10` **User and Role-Based Access Control**

User getPermissions() overridden by Admin, Manager, and Guest.

SAMPLE OUTPUT

```
Admin: Full Access
Manager: Read-Write Access
Guest: Read-Only Access
```

20. ARRAYLIST – Practice Questions

`Question 1` **Create and Display an ArrayList**

Read N integers, store in ArrayList, and display all elements.

SAMPLE INPUT

```
5
10 20 30 40 50
```

SAMPLE OUTPUT

```
[10, 20, 30, 40, 50]
```

Question 2 Add Elements at Specific Indices

Insert element at specific index using add(index, element).

SAMPLE INPUT

```
4
10 20 40 50
30
2
```

SAMPLE OUTPUT

```
[10, 20, 30, 40, 50]
```

Question 3 Remove an Element by Value and Index

Remove element by index using remove(int) and by object value using remove(Object).

SAMPLE INPUT

```
5
10 20 30 40 50
1
30
```

SAMPLE OUTPUT

```
[10, 40, 50]
```

Question 4 Search for an Element in an ArrayList

Check if element exists using contains() and find index using indexOf().

SAMPLE INPUT

```
5
10 20 30 40 50
30
```

SAMPLE OUTPUT

```
Found at index: 2
```

Question 5

Update an Element Using set()

Update element at specific index using set(index, element).

SAMPLE INPUT

```
5
10 20 30 40 50
2
99
```

SAMPLE OUTPUT

```
[10, 20, 99, 40, 50]
```

Question 6

Sort an ArrayList in Ascending and Descending Order

Sort using Collections.sort() and Collections.reverseOrder().

SAMPLE INPUT

```
5
40 10 50 20 30
```

SAMPLE OUTPUT

```
Ascending: [10, 20, 30, 40, 50]
Descending: [50, 40, 30, 20, 10]
```

Question 7

Iterate over an ArrayList

Iterate using for-each loop, standard for loop, and Iterator.

SAMPLE INPUT

```
3
Apple Banana Cherry
```

SAMPLE OUTPUT

```
Apple
Banana
Cherry
```

Question 8 **Reverse an ArrayList**

Reverse list in-place using Collections.reverse().

SAMPLE INPUT

```
5
1 2 3 4 5
```

SAMPLE OUTPUT

```
[5, 4, 3, 2, 1]
```

Question 9 **Remove Duplicates from an ArrayList**

Remove duplicate elements while preserving original insertion order.

SAMPLE INPUT

```
7
10 20 10 30 20 40 50
```

SAMPLE OUTPUT

```
[10, 20, 30, 40, 50]
```

Question 10 **Convert ArrayList to Array and Vice Versa**

Convert ArrayList to Integer[] using toArray() and Array to ArrayList using Arrays.asList().

SAMPLE INPUT

```
3
100 200 300
```

SAMPLE OUTPUT

```
Array Length: 3
ArrayList Size: 3
```

Question 11 Find Maximum and Minimum Elements

Find max and min values using Collections.max() and Collections.min().

SAMPLE INPUT

```
6
45 12 89 23 67 5
```

SAMPLE OUTPUT

```
Maximum: 89
Minimum: 5
```

Question 12 Find Common Elements Between Two ArrayLists

Find intersection of two ArrayLists using retainAll().

SAMPLE INPUT

```
5
1 2 3 4 5
4
3 4 5 6
```

SAMPLE OUTPUT

```
[3, 4, 5]
```

Question 13 Filter Even Numbers from ArrayList

Remove odd numbers keeping only even numbers using removeIf().

SAMPLE INPUT

```
6
1 2 3 4 5 6
```

SAMPLE OUTPUT

```
[2, 4, 6]
```

Question 14 Store Custom Objects in ArrayList

Store Student objects (id, name, marks) in ArrayList and display formatted records.

SAMPLE INPUT

```
2
101 Rahul 85
102 Sneha 92
```

SAMPLE OUTPUT

```
ID: 101, Name: Rahul, Marks: 85.0
ID: 102, Name: Sneha, Marks: 92.0
```

Question 15 Sort Custom Objects Using Comparator

Sort list of Student objects by marks descending using Comparator.

SAMPLE INPUT

```
3
101 Rahul 75
102 Sneha 95
103 Amit 88
```

SAMPLE OUTPUT

```
Sneha (95.0)
Amit (88.0)
Rahul (75.0)
```

21. HASHSET – Practice Questions

`Question 1` **Add and Display Unique Elements**

Read elements into HashSet and demonstrate automatic duplicate elimination.

SAMPLE INPUT

```
6
10 20 10 30 20 40
```

SAMPLE OUTPUT

```
Unique Elements Count: 4
```

`Question 2` **Check if an Element Exists**

Check presence of given target element using contains().

SAMPLE INPUT

```
4
Apple Banana Cherry Date
Banana
```

SAMPLE OUTPUT

```
Element Exists
```

`Question 3` **Remove an Element from HashSet**

Remove specified element using remove().

SAMPLE INPUT

```
4
10 20 30 40
20
```

SAMPLE OUTPUT

```
Element Removed Successfully
```

Question 4 Find Union of Two Sets

Calculate union of two sets using addAll().

SAMPLE INPUT

```
3
1 2 3
3
3 4 5
```

SAMPLE OUTPUT

```
Union Count: 5
```

Question 5 Find Intersection of Two Sets

Calculate intersection of two sets using retainAll().

SAMPLE INPUT

```
4
1 2 3 4
3
3 4 5
```

SAMPLE OUTPUT

```
Intersection: [3, 4]
```

Question 6 Find Difference Between Two Sets

Calculate set difference (SetA - SetB) using removeAll().

SAMPLE INPUT

```
4
1 2 3 4
2
3 4
```

SAMPLE OUTPUT

```
Difference: [1, 2]
```

Question 7 Convert HashSet to ArrayList and Sort

Convert unordered HashSet to ArrayList and sort using Collections.sort().

SAMPLE INPUT

```
5
40 10 50 20 30
```

SAMPLE OUTPUT

```
Sorted List: [10, 20, 30, 40, 50]
```

Question 8 Check SubSet Relationship

Check whether SetB is a subset of SetA using containsAll().

SAMPLE INPUT

```
5
1 2 3 4 5
3
2 3 4
```

SAMPLE OUTPUT

```
Is Subset: true
```

`Question 9` **Find First Duplicate in an Array Using HashSet**

Identify first repeated element in an array using HashSet tracking.

SAMPLE INPUT

```
6
2 5 1 2 3 5
```

SAMPLE OUTPUT

```
First Duplicate: 2
```

`Question 10` **Remove Duplicate Words from a Sentence**

Print unique words from a sentence preserving order using LinkedHashSet.

SAMPLE INPUT

```
Java is great and Java is fast
```

SAMPLE OUTPUT

```
Java is great and fast
```

22. HASHMAP – Practice Questions

Question 1 Store and Display Key-Value Pairs

Store roll number (key) and student name (value) in HashMap and display all entries.

SAMPLE INPUT

```
3
101 Rahul
102 Sneha
103 Amit
```

SAMPLE OUTPUT

```
101 -> Rahul
102 -> Sneha
103 -> Amit
```

Question 2 Check if a Key or Value Exists

Check key presence using containsKey() and value presence using containsValue().

SAMPLE INPUT

```
102
Rahul
```

SAMPLE OUTPUT

```
Key Exists: true
Value Exists: true
```

Question 3 Update and Remove Entries in HashMap

Update value using put() / replace() and remove entry using remove(key).

SAMPLE INPUT

```
101 Vikram
102
```

SAMPLE OUTPUT

```
Updated Map: {101=Vikram, 103=Amit}
```

Question 4 Count Character Frequencies in a String

Count frequency of each character in string using HashMap.

SAMPLE INPUT

```
programming
```

SAMPLE OUTPUT

```
p : 1
r : 2
o : 1
g : 2
a : 1
m : 2
i : 1
n : 1
```

Question 5 Count Word Frequencies in a Sentence

Count frequency of each word in a sentence ignoring case.

SAMPLE INPUT

```
Java is fun and Java is powerful
```

SAMPLE OUTPUT

```
java : 2
is : 2
fun : 1
and : 1
powerful : 1
```

Question 6 Iterate over a HashMap

Iterate using entrySet(), keySet(), values(), and forEach() lambda.

SAMPLE OUTPUT

```
Key: 101, Value: Rahul
Key: 102, Value: Sneha
```

Question 7 Sort HashMap by Keys

Sort HashMap entries by key by copying into TreeMap.

SAMPLE INPUT

```
3
103 Amit
101 Rahul
102 Sneha
```

SAMPLE OUTPUT

```
101 -> Rahul
102 -> Sneha
103 -> Amit
```

Question 8 Sort HashMap by Values

Sort HashMap entries by value using List of Map.Entry and Comparator.

SAMPLE INPUT

```
3
Rahul 85
Sneha 95
Amit 78
```

SAMPLE OUTPUT

```
Amit -> 78
Rahul -> 85
Sneha -> 95
```

`Question 9` **Find the Most Frequent Element in an Array**

Find element with maximum frequency in integer array using HashMap.

SAMPLE INPUT

```
8
1 3 2 1 4 1 3 1
```

SAMPLE OUTPUT

```
Most Frequent Element: 1 (Frequenc
y: 4)
```

`Question 10` **Find Two Numbers that Sum to Target (Two Sum)**

Find indices of two numbers in array that add up to target using HashMap in O(N).

SAMPLE INPUT

```
4
2 7 11 15
9
```

SAMPLE OUTPUT

```
Indices: [0, 1]
```

23. SORTING ALGORITHMS – Practice Questions

Question 1 Implement Bubble Sort Algorithm

Sort integer array in ascending order using Bubble Sort algorithm.

SAMPLE INPUT

```
5
64 34 25 12 22
```

SAMPLE OUTPUT

```
Sorted Array: 12 22 25 34 64
```

Question 2 Implement Selection Sort Algorithm

Sort integer array in ascending order using Selection Sort algorithm.

SAMPLE INPUT

```
5
64 25 12 22 11
```

SAMPLE OUTPUT

```
Sorted Array: 11 12 22 25 64
```

Question 3 Implement Insertion Sort Algorithm

Sort integer array in ascending order using Insertion Sort algorithm.

SAMPLE INPUT

```
6
12 11 13 5 6 7
```

SAMPLE OUTPUT

```
Sorted Array: 5 6 7 11 12 13
```

Question 4

Implement Merge Sort Algorithm

Sort integer array using divide-and-conquer Merge Sort algorithm.

SAMPLE INPUT

```
7
38 27 43 3 9 82 10
```

SAMPLE OUTPUT

```
Sorted Array: 3 9 10 27 38 43 82
```

Question 5

Implement Quick Sort Algorithm

Sort integer array using Quick Sort with last element as pivot.

SAMPLE INPUT

```
6
10 80 30 90 40 50
```

SAMPLE OUTPUT

```
Sorted Array: 10 30 40 50 80 90
```

24. SEARCHING ALGORITHMS – Practice Questions

`Question 1`

Implement Linear Search Algorithm

Search for target value by checking each element sequentially. Return index or -1.

SAMPLE INPUT

```
5
10 20 30 40 50
30
```

SAMPLE OUTPUT

```
Element found at index 2
```

`Question 2`

Implement Binary Search (Iterative)

Iterative binary search on sorted integer array. Return index or -1.

SAMPLE INPUT

```
6
10 20 30 40 50 60
40
```

SAMPLE OUTPUT

```
Element found at index 3
```

`Question 3`

Implement Binary Search (Recursive)

Recursive binary search on sorted integer array. Return index or -1.

SAMPLE INPUT

```
6
10 20 30 40 50 60
50
```

SAMPLE OUTPUT

```
Element found at index 4
```

`Question 4` **Find First and Last Occurrence in a Sorted Array**

Find first and last occurrence indices of target in sorted array containing duplicates.

SAMPLE INPUT

```
7
1 2 2 2 3 4 5
2
```

SAMPLE OUTPUT

```
First Occurrence: Index 1
Last Occurrence: Index 3
```

25. JAVA MINI PROJECTS

Development Guidelines for Mini Projects

- Use modular design: Separate domain models, management services, and user interface components into distinct classes.
- Implement robust error handling for invalid user inputs using try-catch blocks and loops.
- Store data in dynamic collection structures (ArrayList, HashMap) to support flexible operations.

Project 1 Student Grade Management System

Requirements: Build an interactive console application to manage student records. Support adding students, listing records, calculating subject averages, assigning letter grades (A/B/C/D/F), finding top scorer, and searching student by ID.

CONSOLE MENU

```
1. Add Student
2. Display All Students
3. Search Student by ID
4. Find Top Scorer
5. Exit
```

SAMPLE RUN

```
Added Student: Rahul (ID: 101, Avg:
88.5, Grade: A)
Top Scorer: Sneha (ID: 102, Avg: 9
4.0, Grade: A)
```

Project 2 Bank Account Management System

Requirements: Implement account management supporting Savings and Current accounts. Support account creation, deposit, withdrawal with validation, balance lookup, and transaction history logging.

CONSOLE MENU

```
1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. View Statement
6. Exit
```

SAMPLE RUN

```
Deposited: Rs. 5000. Balance: Rs. 1
5000
Withdrew: Rs. 2000. Balance: Rs. 13
000
```

`Project 3` **Library Book Issue and Return System**

Requirements: Console system to manage library operations. Support adding books, searching by title/author, issuing books, returning books, and tracking issued status per student.

CONSOLE MENU

```
1. Add Book
2. Display Books
3. Issue Book
4. Return Book
5. Search Book
6. Exit
```

SAMPLE RUN

```
Issued 'Clean Code' to Student ID 1
01
Book 'Clean Code' successfully retu
rned
```

`Project 4` **ATM Simulation Machine**

Requirements: Simulate ATM machine with PIN authentication (max 3 attempts). Options: PIN change, cash withdrawal (multiples of 100/500), fast cash, deposit, mini-statement.

CONSOLE MENU

```
1. Balance Inquiry
2. Cash Withdrawal
3. Deposit
4. Change PIN
5. Mini Statement
6. Exit
```

SAMPLE RUN

```
PIN Verified. Welcome!
Please collect cash: Rs. 2000
```

Project 5 Inventory Management System

Requirements: Console inventory tracker for retail store. Track product ID, name, quantity, unit price. Support stock update, low stock alert (<10 items), and total inventory valuation.

CONSOLE MENU

```
1. Add Product
2. Update Stock
3. Display Inventory
4. Low Stock Alert
5. Calculate Total Value
6. Exit
```

SAMPLE RUN

```
Alert: Product 'Mouse' stock low (5 units)
Total Inventory Valuation: Rs. 1,45,000
```

Project 6 Simple Quiz Application

Requirements: Interactive multiple-choice quiz engine. Present questions, accept options, score user, display result report, and high scores leaderboard.

CONSOLE DISPLAY

```
Q1: What is default value of int in Java?
A) 0  B) null  C) 1  D) Undefined
Your Choice: A
```

SAMPLE RUN

```
Correct Answer! +1 Mark
Quiz Complete! Score: 8/10 (80%)
```